ALY6010 [20215]
Prob. Theory & Intro. Statistics

Northeastern

M4: Comparison Tests

Richard He
College of Professional Studies
Tuesday, Winter B 2025

Topics Covered in this Course

- **Probability Distribution (Week 1)**
- **Confidence Intervals (Week 2)**
- **Hypothesis Testing (Week 3)**
- **Comparison Tests (Week 4)**
- Correlation & Regression Part I (Week 5)
- Correlation & Regression Part II (Week 6)

Today's Agenda

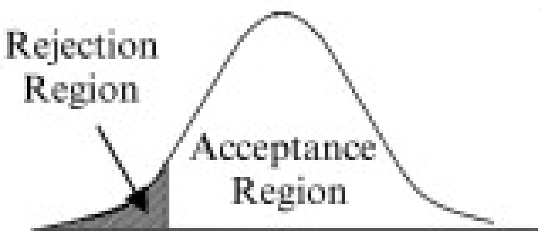
By the end of this module, you should be able to:

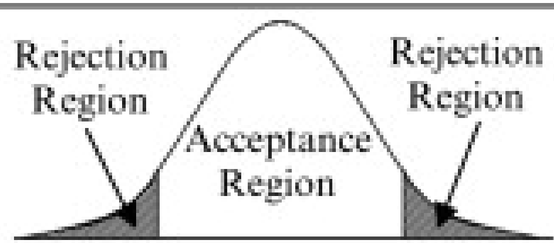
- Conduct two sample hypothesis testing
- Conduct hypothesis testing for two independent samples
- Conduct subgroup hypothesis testing
- Use R to create charts by groups

Recap

Types of Tailed Test

One-Tailed Test (Left Tail)	Two-Tailed Test	One-Tailed Test (Right Tail)
$H_0: \mu_X = \mu_0$ $H_1: \mu_X < \mu_0$	$H_0: \mu_X = \mu_0$ $H_1: \mu_X \neq \mu_0$	$H_0: \mu_X = \mu_0$ $H_1: \mu_X > \mu_0$





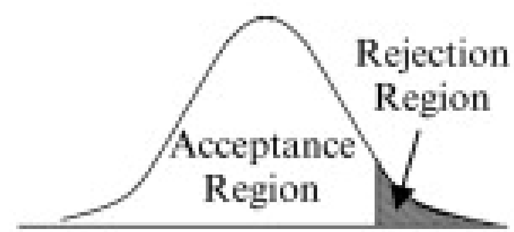


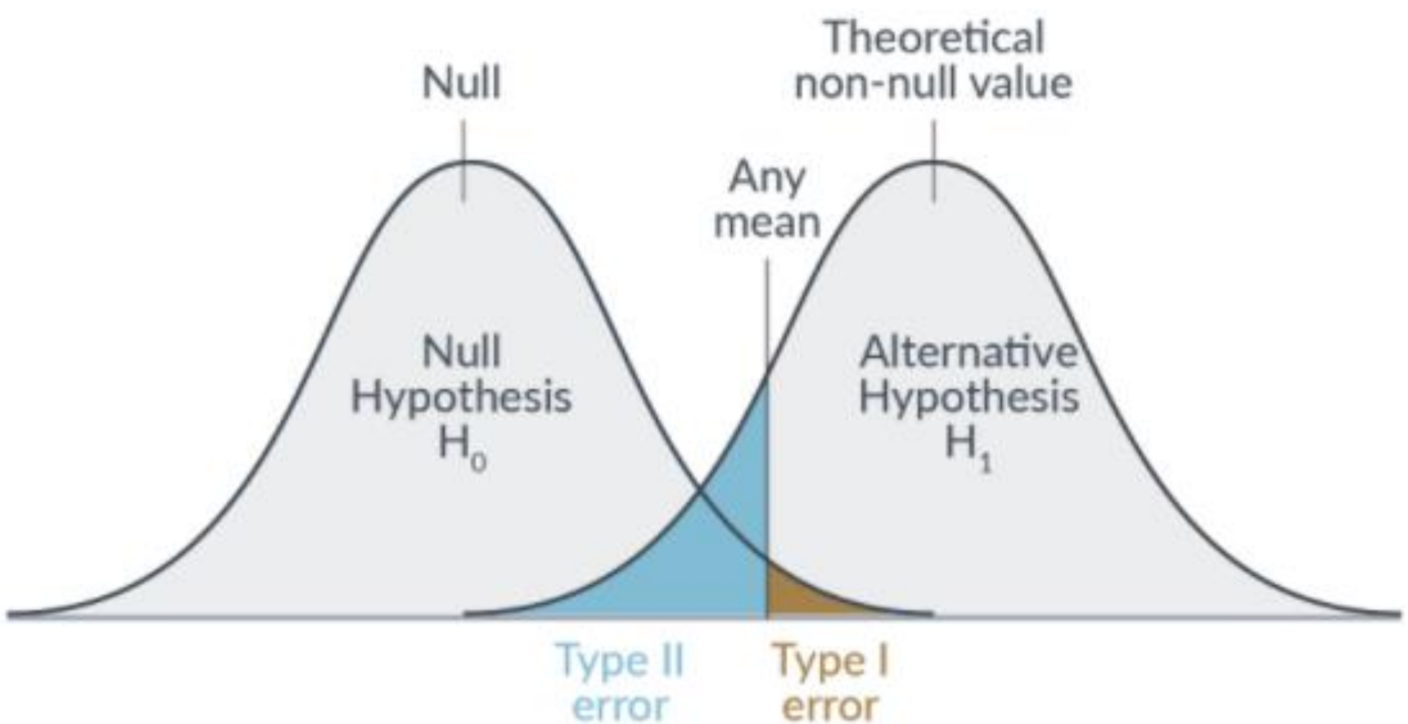
Image source: www.sciencedirect.com/topics/mathematics/tailed-test

Methods of Decision Making

- Comparing test statistic to critical value
- Comparing p-value to alpha(significant level)
- Using confidence intervals

Type I & II Errors

	Description	
	Accept H_0	Reject H_0
H_0 (true)	Correct decision	Type I error (α error)
H_0 (false)	Type II error (β error)	Correct decision



Types of Hypothesis Testing

- Z-test for mean
- T-test for mean
- Z-test for proportion
- Chi-square test for variance

Lecture Part I
Comparison Tests

Motivation for Comparison Tests

- The needs to compare statistics between two samples
  - The average lifetimes of two different brands of tires
  - Two different brands of fertilizer might be tested to see whether one is better than the other for growing plants
  - Or two brands of cough syrup might be tested to see whether one is more effective than the other

General Assumptions

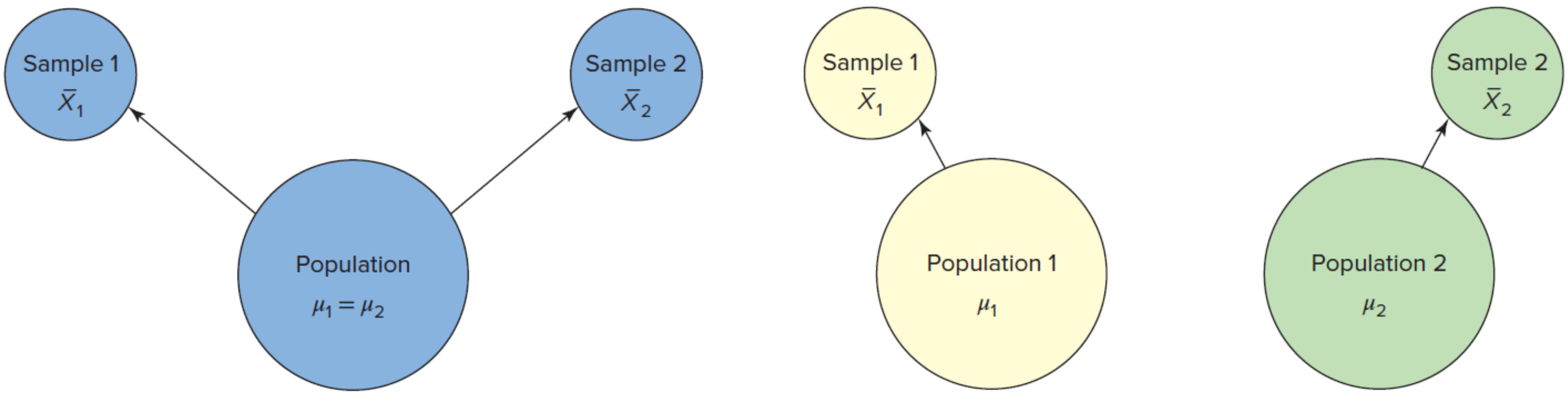
- Samples are random
- The observation should be **independent**
- The population should be **normally distributed**

Types of Comparison Tests

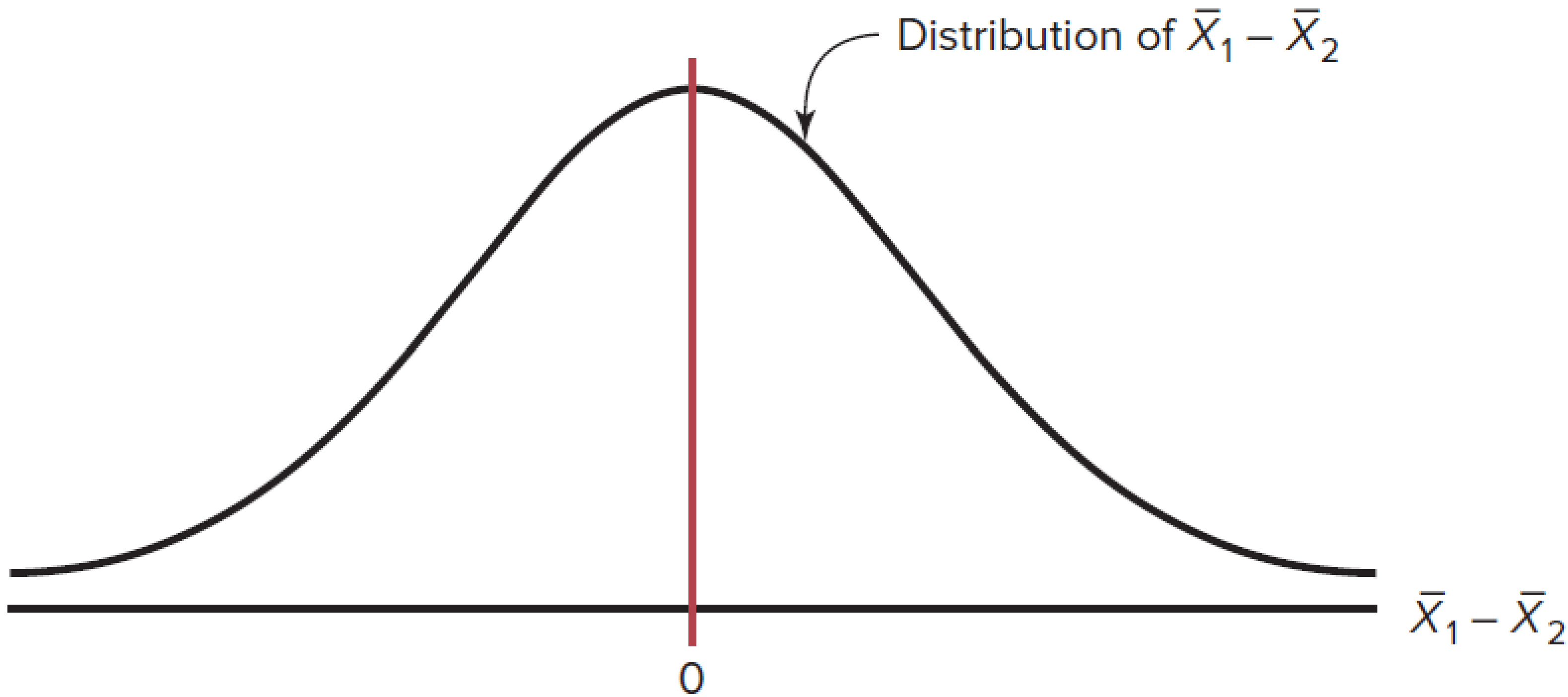
- Z-Test for Two Means
- T-Test for Two Means (Independent)
- T-Test for Two Means (Dependent)
- Z-Test for Two Proportions
- F-Test for Two Variances

Z-Test for Two Means

Scenarios of Comparing Means



Difference of Two Means



$$\sigma^2_{\bar{X}_1 - \bar{X}_2} = \sigma^2_{\bar{X}_1} + \sigma^2_{\bar{X}_2}$$

$$\sigma^2_{\bar{X}_1} = \frac{\sigma_1^2}{n_1} \quad \text{and} \quad \sigma^2_{\bar{X}_2} = \frac{\sigma_2^2}{n_2}$$

Mean & Variance of Sample Mean

- Mean of the sample mean

$$E[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \mu$$

- Variance of the sample mean

$$Var[\bar{X}] = Var\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{\sigma^2}{n}$$

Variance of Two Sample Means

- Mean of two sample means

$$E[\bar{X}_1 - \bar{X}_2] = \mu_1 - \mu_2$$

- Variance of two sample means

$$Var[\bar{X}_1 - \bar{X}_2] = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

Z Score & Confidence Interval

$$z = \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$$

$$(\overline{X}_1 - \overline{X}_2) - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\overline{X}_1 - \overline{X}_2) + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Hypotheses for Z-Test

$$H_0\colon \mu_1 = \mu_2 \qquad H_0\colon \mu_1 - \mu_2 = 0$$
$$H_1\colon \mu_1 \neq \mu_2 \qquad H_1\colon \mu_1 - \mu_2 \neq 0$$

Right-tailed			Left-tailed		
$H_0\colon \mu_1 = \mu_2$ $H_1\colon \mu_1 > \mu_2$	or	$H_0\colon \mu_1 - \mu_2 = 0$ $H_1\colon \mu_1 - \mu_2 > 0$	$H_0\colon \mu_1 = \mu_2$ $H_1\colon \mu_1 < \mu_2$	or	$H_0\colon \mu_1 - \mu_2 = 0$ $H_1\colon \mu_1 - \mu_2 < 0$

Ex 9.1 Leisure Time

Q: A study using two random samples of **35 people** each found that the average amount of time those in the age group of **26–35 years** spent per week on leisure activities was **39.6 hours**, and those in the age group of **46–55 years** spent **35.4 hours**.

Assume that the population standard deviation for those in the first age group found by previous studies is **6.3 hours**, and the population standard deviation of those in the second group found by previous studies was **5.8 hours**.

At $\alpha = 0.05$, can it be concluded that there is a significant difference in the average times each group spends on leisure activities?

Ex 9.1 Solution

Step 1 State the hypotheses and identify the claim.

$$H_0\colon \mu_1 = \mu_2 \qquad \text{and} \qquad H_1\colon \mu_1 \neq \mu_2 \text{ (claim)}$$

Step 2 Find the critical values. Since $\alpha = 0.05$, the critical values are $+1.96$ and -1.96.

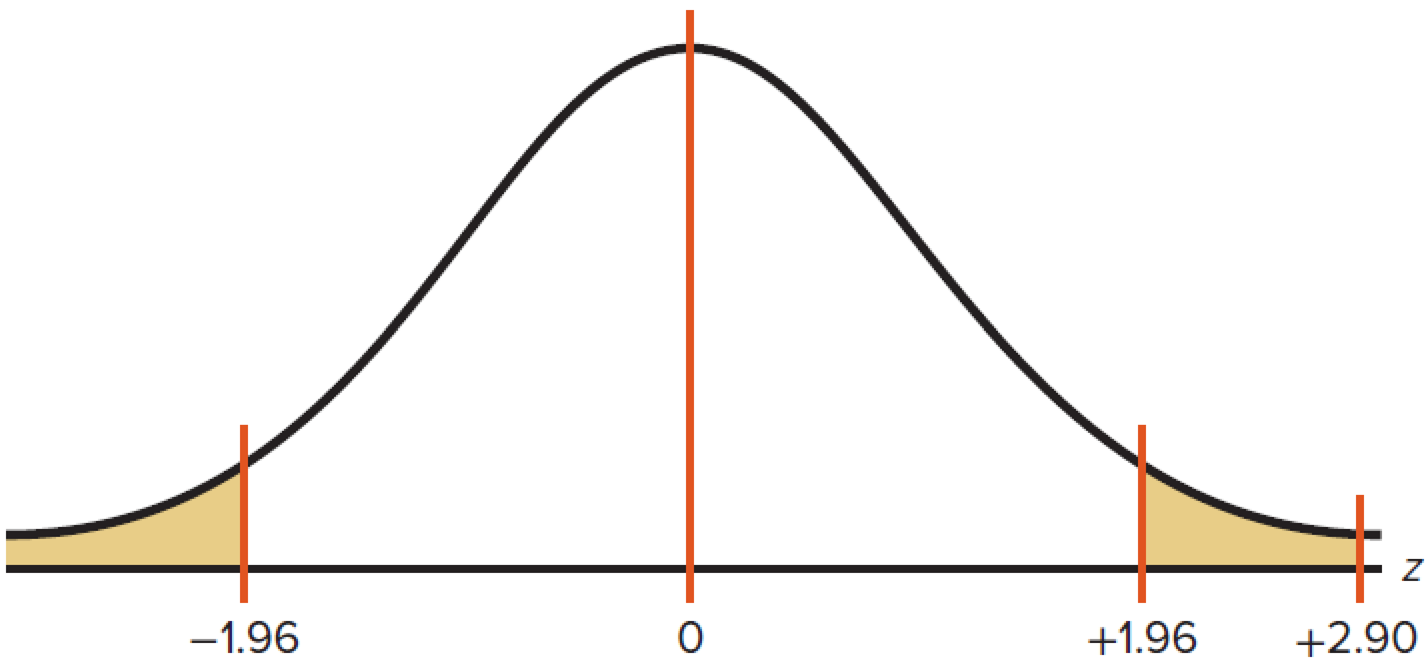
Step 3 Compute the test value.

$$z = \frac{(\overline{X}_1 - \overline{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}} = \frac{(39.6 - 35.4) - 0}{\sqrt{\dfrac{6.3^2}{35} + \dfrac{5.8^2}{35}}} = \frac{4.2}{1.447} = 2.90$$

Step 4 Make the decision. Reject the null hypothesis at $\alpha = 0.05$ since $2.90 > 1.96$.

Step 5 Summarize the results. There is enough evidence to support the claim that the means are not equal. That is, the average of the times spent on leisure activities is different for the groups.

Ex 9.1 Critical Value & P-Value




```
> pnorm(2.90, lower.tail=FALSE)
[1] 0.001865813
```

Ex 9.1 95% Confidence Interval

$$(\overline{X}_1 - \overline{X}_2) - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\overline{X}_1 - \overline{X}_2) + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$(39.6 - 35.4) - 1.96\sqrt{\frac{6.3^2}{35} + \frac{5.8^2}{35}} < \mu_1 - \mu_2 < (39.6 - 35.4) + 1.96\sqrt{\frac{6.3^2}{35} + \frac{5.8^2}{35}}$$

$$4.2 - 2.8 < \mu_1 - \mu_2 < 4.2 + 2.8$$

$$1.4 < \mu_1 - \mu_2 < 7.0$$

Since the confidence interval does not contain zero, the decision is to reject the null hypothesis, which agrees with the previous result.

T-Test for Two Means (Independent Samples)

T Score & Confidence Interval

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$$

degrees of freedom are equal to the smaller of $n_1 - 1$ or $n_2 - 1$

$$(\bar{X}_1 - \bar{X}_2) - t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{X}_1 - \bar{X}_2) + t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

d.f. = smaller value of $n_1 - 1$ or $n_2 - 1$

Ex 9.4 Work Absences

Q: A study was done to see if there is a difference between the number of sick days men take and the number of sick days women take.

A random sample of 9 men found that the mean of the number of sick days taken was 5.5. The standard deviation of the sample was 1.23. A random sample of 7 women found that the mean was 4.3 days and a standard deviation of 1.19 days.

At $\alpha = 0.05$, can it be concluded that there is a difference in the means?

Ex 9.4 Solution

Step 1 State the hypotheses and identify the claim.

$$H_0\colon \mu_1 = \mu_2 \quad \text{and} \quad H_1\colon \mu_1 \neq \mu_2 \text{ (claim)}$$

Step 2 Find the critical values.

```
> qt(0.025, df=7-1)
[1] -2.446912
> qt(0.975, df=7-1)
[1] 2.446912
```

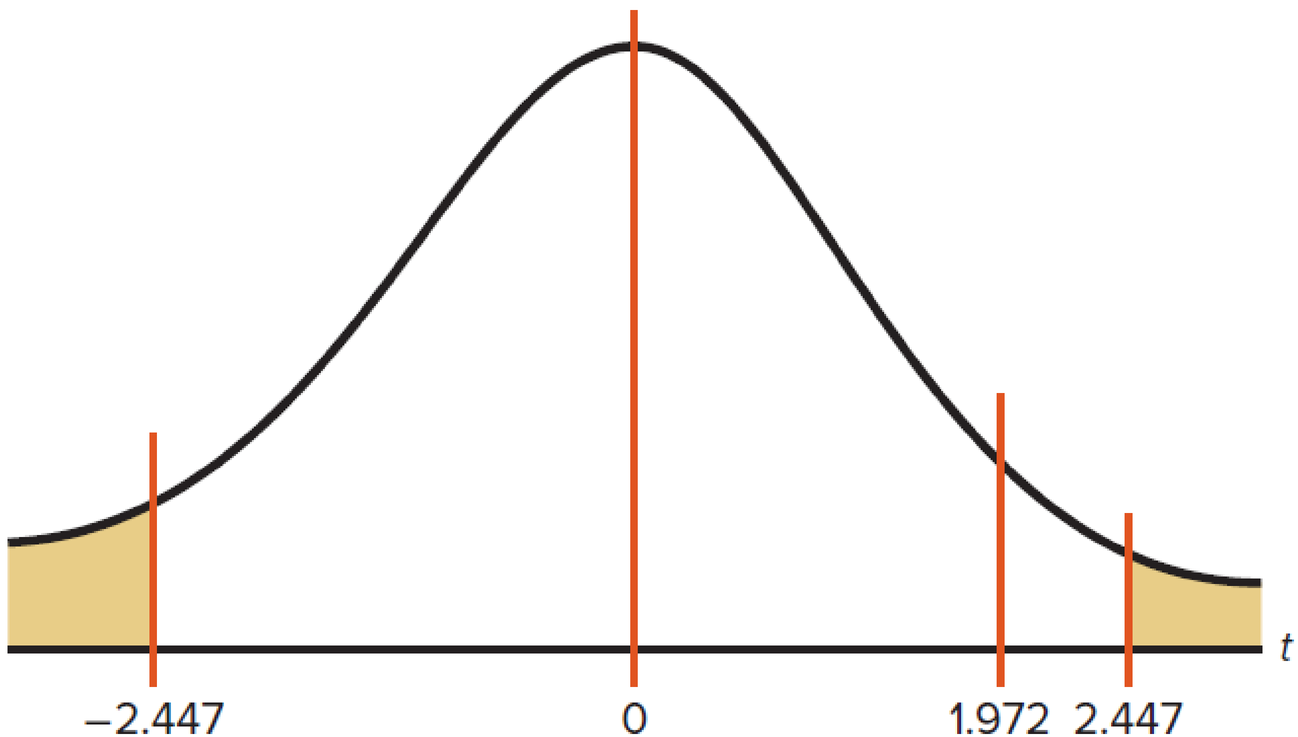
Step 3 Compute the test value.

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(5.5 - 4.3) - 0}{\sqrt{\frac{1.23^2}{9} + \frac{1.19^2}{7}}} = 1.972$$

Step 4 Make the decision. Do not reject the null hypothesis since $1.972 < 2.447$.

Step 5 Summarize the results. There is not enough evidence to support the claim that the means are different.

Ex 9.4 Critical Value & P-Value




```
> pt(1.972, df=7-1, lower.tail = FALSE)
[1] 0.04804174
```

Ex 9.4 95% Confidence Interval

$$(\overline{X}_1 - \overline{X}_2) - t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\overline{X}_1 - \overline{X}_2) + t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$(5.5 - 4.3) - 2.447\sqrt{\frac{1.23^2}{9} + \frac{1.19^2}{7}} < \mu_1 - \mu_2 < (5.5 - 4.3) + 2.447\sqrt{\frac{1.23^2}{9} + \frac{1.19^2}{7}}$$

$$1.2 - 1.489 < \mu_1 - \mu_2 < 1.2 + 1.489$$

$$-0.289 < \mu_1 - \mu_2 < 2.689$$

Since 0 is contained in the interval, there is not enough evidence to support the claim that the means are different.

T-Test for Two Means (Dependent Samples)

Dependent Samples

- Also called **matched-pair** samples

- Same subjects are used in a pre-post situation
  - Losing weight program: pre-weight vs post-weight
  - Reaction time & drug: pretest vs posttest

T Score & Confidence Interval

$$t = \frac{\overline{D} - \mu_D}{s_D / \sqrt{n}} \quad \text{with d.f.} = n - 1$$

$$\overline{D} = \frac{\Sigma D}{n} \qquad \mu_D = 0 \qquad s_{\overline{D}} = \frac{s_D}{\sqrt{n}} \qquad s_D = \sqrt{\frac{n\Sigma D^2 - (\Sigma D)^2}{n(n-1)}}$$

$$\overline{D} - t_{\alpha/2}\frac{s_D}{\sqrt{n}} < \mu_D < \overline{D} + t_{\alpha/2}\frac{s_D}{\sqrt{n}}$$

$$\text{d.f.} = n - 1$$

Computing the Test Value

a. Make a table, as shown.

X_1	X_2	A $D = X_1 - X_2$	B $D^2 = (X_1 - X_2)^2$
⋮	⋮	$\Sigma D =$ ______	$\Sigma D^2 =$ ______

b. Find the differences and place the results in column A.

$$D = X_1 - X_2$$

c. Find the mean of the differences.

$$\overline{D} = \frac{\Sigma D}{n}$$

d. Square the differences and place the results in column B. Complete the table.

$$D^2 = (X_1 - X_2)^2$$

e. Find the standard deviation of the differences.

$$s_D = \sqrt{\frac{n\Sigma D^2 - (\Sigma D)^2}{n(n-1)}}$$

f. Find the test value.

$$t = \frac{\overline{D} - \mu_D}{s_D/\sqrt{n}} \qquad \text{with d.f.} = n - 1$$

Hypotheses for T-Test

Two-tailed	Left-tailed	Right-tailed
H_0: $\mu_D = 0$ H_1: $\mu_D \neq 0$	H_0: $\mu_D = 0$ H_1: $\mu_D < 0$	H_0: $\mu_D = 0$ H_1: $\mu_D > 0$

μ_D is the symbol for the expected mean of the difference of the matched pairs

Ex 9.7 Cholesterol Levels

Q: A dietitian wishes to see if a person's cholesterol level will change if the diet is supplemented

by a certain mineral. **Six** randomly selected subjects were pretested, and then they took the mineral supplement for a 6-week period.

Subject	1	2	3	4	5	6
Before (X_1)	210	235	208	190	172	244
After (X_2)	190	170	210	188	173	228

```
> qt(0.05, df=6-1)
[1] -2.015048
> qt(0.05, df=6-1, lower.tail = FALSE)
[1] 2.015048
```

Ex 9.7 Solution

Step 1 State the hypotheses and identify the claim. If the diet is effective, the before cholesterol levels should be different from the after levels.

$$H_0\colon \mu_D = 0 \qquad \text{and} \qquad H_1\colon \mu_D \neq 0 \text{ (claim)}$$

Step 2 Find the critical value. The degrees of freedom are $6 - 1 = 5$. At $\alpha = 0.10$, the critical values are ± 2.015.

$$t = \frac{\overline{D} - \mu_D}{s_D/\sqrt{n}} = \frac{16.7 - 0}{25.4/\sqrt{6}} = 1.610$$

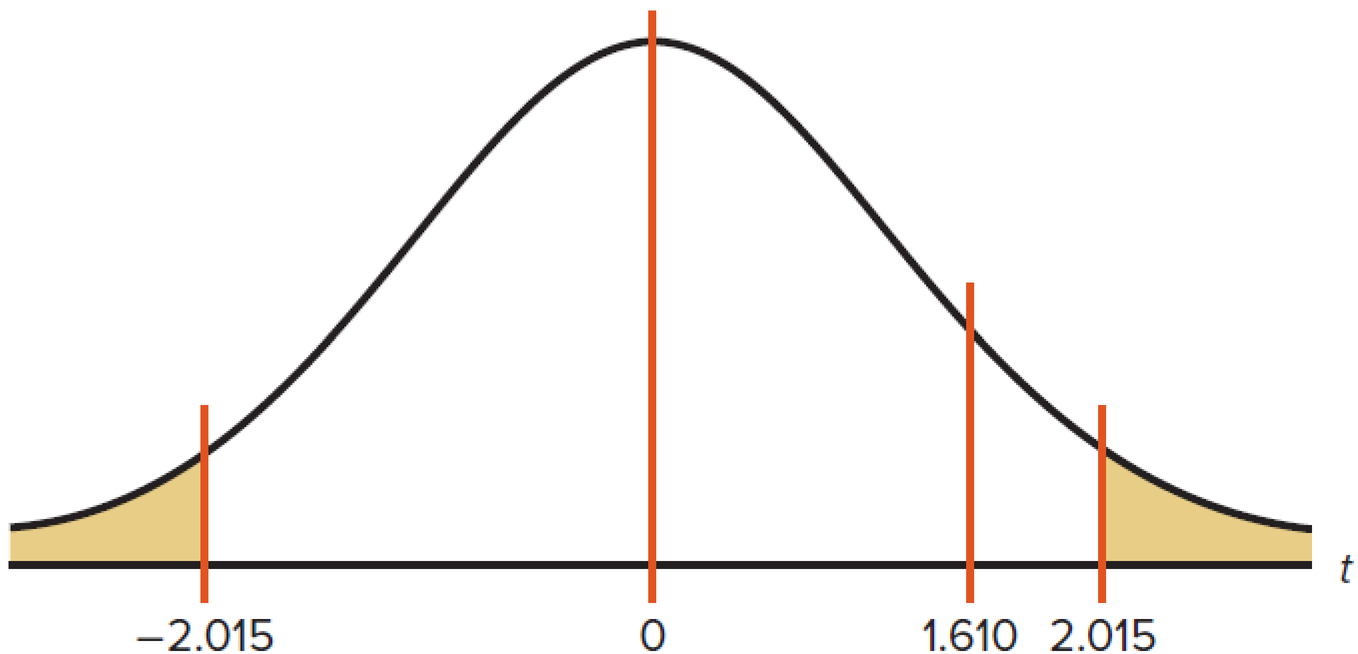
Step 3 Compute the test value.

X_1	X_2	A $D = X_1 - X_2$	B $D^2 = (X_1 - X_2)^2$
⋮	⋮	$\Sigma D =$ 100	$\Sigma D^2 =$ 4890

Step 4 Make the decision. The decision is to not reject the null hypothesis

Step 5 Summarize the results. There is not enough evidence to support the claim that the mineral changes a person's cholesterol level.

Ex 9.7 Critical Value & P-Value




```
> pt(1.61, df=6-1, lower.tail = FALSE)
[1] 0.08415637
> pt(2.015, df=6-1, lower.tail = FALSE)
[1] 0.05000309
```

$$s_D = \sqrt{\frac{n\Sigma D^2 - (\Sigma D)^2}{n(n-1)}}$$

Ex 9.7 90% Confidence Interval

$$\overline{D} - t_{\alpha/2}\frac{s_D}{\sqrt{n}} < \mu_D < \overline{D} + t_{\alpha/2}\frac{s_D}{\sqrt{n}}$$

$$16.7 - 2.015 \cdot \frac{25.4}{\sqrt{6}} < \mu_D < 16.7 + 2.015 \cdot \frac{25.4}{\sqrt{6}}$$

$$16.7 - 20.89 < \mu_D < 16.7 + 20.89$$

$$-4.19 < \mu_D < 37.59$$

$$-4.2 < \mu_D < 37.6$$

Since 0 is contained in the interval, the decision is to not reject the null hypothesis

Z-Test for Two Proportions

Variance of Bernoulli R.V.

- Expectation

$$E[X] = 1 * p + 0 * p = p$$

- Variance

$$Var(X) = E[X^2] - (E[X])^2$$

$$= 1^2 * p - p^2 = pq$$

Variance of Sample Proportion

- Sample proportion

$$p = \frac{X}{n}$$

- Variance of the sample mean

$$Var(p) = Var\left(\frac{X}{n}\right) = \frac{1}{n^2}(npq)$$

$$= \frac{pq}{n} = \frac{p(1-p)}{n}$$

Z Score

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}\,\bar{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2} \qquad \hat{p}_1 = \frac{X_1}{n_1}$$

$$\bar{q} = 1 - \bar{p} \qquad \hat{p}_2 = \frac{X_2}{n_2}$$

$$\bar{p} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}$$

$$\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{\bar{p}\,\bar{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

Ex 9.10 Vaccination Rates

Q: In the nursing home study mentioned in the chapter-opening Statistics Today, the researchers found that 12 out of 34 randomly selected small nursing homes had a resident vaccination rate of less than 80%, while 17 out of 24 randomly selected large nursing homes had a vaccination rate of less than 80%. At $\alpha = 0.05$, test the claim that there is **no difference in the proportions** of the small and large nursing homes with a resident vaccination rate of less than 80%.

Ex 9.10 Solution

Step 1 State the hypotheses and identify the claim.

$$H_0\colon p_1 = p_2 \text{ (claim)} \qquad \text{and} \qquad H_1\colon p_1 \neq p_2$$

Step 2 Find the critical values. Since $\alpha = 0.05$, the critical values are $+1.96$ and -1.96.

Step 3 Compute the test value.

$$\hat{p}_1 = \frac{X_1}{n_1} = \frac{12}{34} = 0.35 \quad \text{and} \quad \hat{p}_2 = \frac{X_2}{n_2} = \frac{17}{24} = 0.71$$

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{12 + 17}{34 + 24} = \frac{29}{58} = 0.5$$

$$\bar{q} = 1 - \bar{p} = 1 - 0.5 = 0.5$$

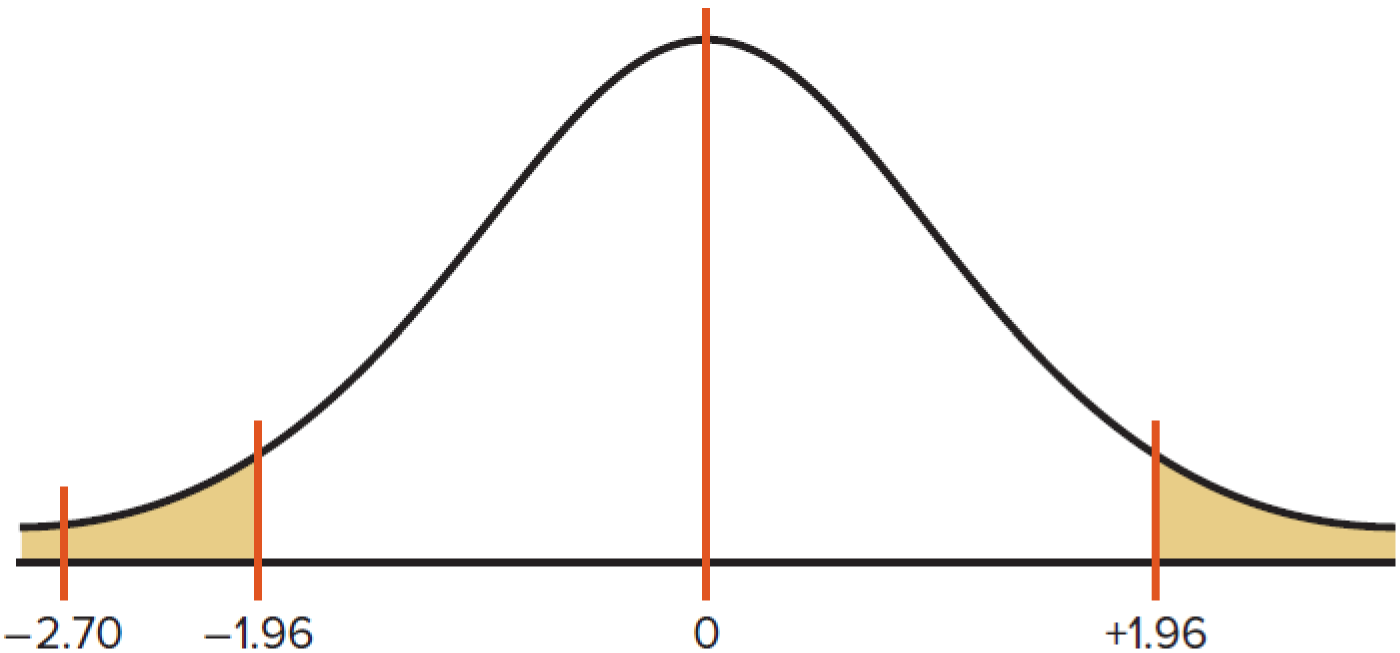
$$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}\,\bar{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$= \frac{(0.35 - 0.71) - 0}{\sqrt{(0.5)(0.5)\left(\frac{1}{34} + \frac{1}{24}\right)}} = \frac{-0.36}{0.1333} = -2.70$$

Step 4 Make the decision. Reject the null hypothesis, since $-2.70 < -1.96$

Step 5 Summarize the results. There is enough evidence to reject the claim

Ex 9.10 Critical Value & P-Value




```
> pnorm(-2.70)
[1] 0.003466974
> pnorm(-1.96)
[1] 0.0249979
```

F-Test for Two Variances

F Distribution

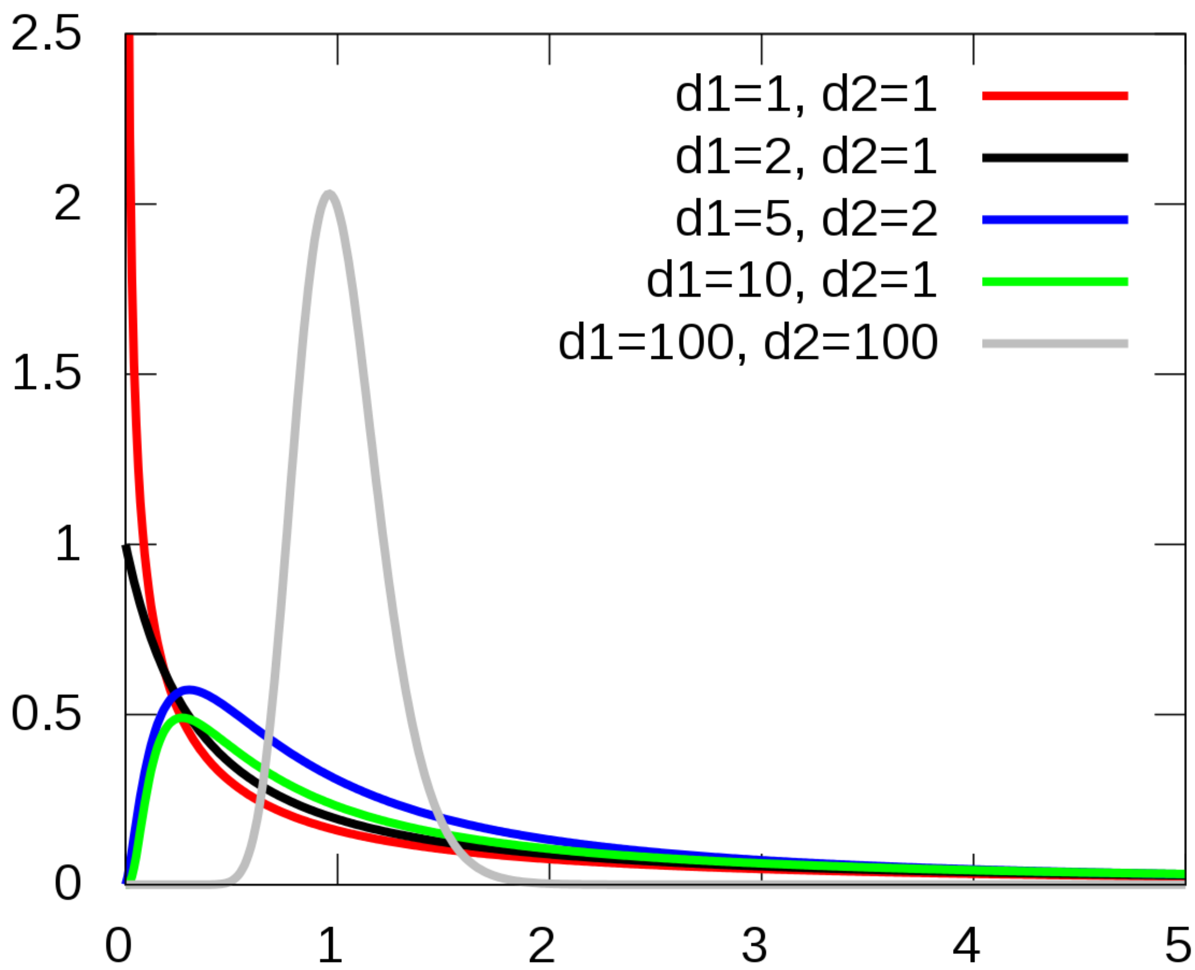


Image source: en.wikipedia.org/wiki/F-distribution

F Test

- Two degree of freedom
  - Numerator: $d.f.N = n_1 - 1$
  - Denominator: $d.f.D = n_2 - 1$
- F score is greater than 1
  - The larger variance: numerator
  - The smaller variance: denominator

$$F = \frac{s_1^2}{s_2^2} \qquad s_1^2 \geq s_2^2$$

Hypotheses for F-Test

Right-tailed	Left-tailed	Two-tailed
$H_0: \sigma_1^2 = \sigma_2^2$ $H_1: \sigma_1^2 > \sigma_2^2$	$H_0: \sigma_1^2 = \sigma_2^2$ $H_1: \sigma_1^2 < \sigma_2^2$	$H_0: \sigma_1^2 = \sigma_2^2$ $H_1: \sigma_1^2 \neq \sigma_2^2$

Ex 9.14 Heart Rates of Smokers

Q: A medical researcher wishes to see whether the variance of the heart rates (in beats per minute) of smokers is different from the variance of heart rates of people who do not smoke. Two samples are selected, and the data are shown. Using α = 0.05, is there enough evidence to support the claim? Assume the variable is normally distributed.

Smokers	Nonsmokers
$n_1 = 26$ $s_1^2 = 36$	$n_2 = 18$ $s_2^2 = 10$

Ex 9.14 Solution

Step 1 State the hypotheses and identify the claim.

$$H_0: \sigma_1^2 = \sigma_2^2 \qquad \text{and} \qquad H_1: \sigma_1^2 \neq \sigma_2^2 \text{ (claim)}$$

Step 2 Find the critical value.

```
> qf(p=0.975, df1=26-1, df2=18-1)
[1] 2.548419
> qf(p=0.025, df1=26-1, df2=18-1)
[1] 0.4237533
```

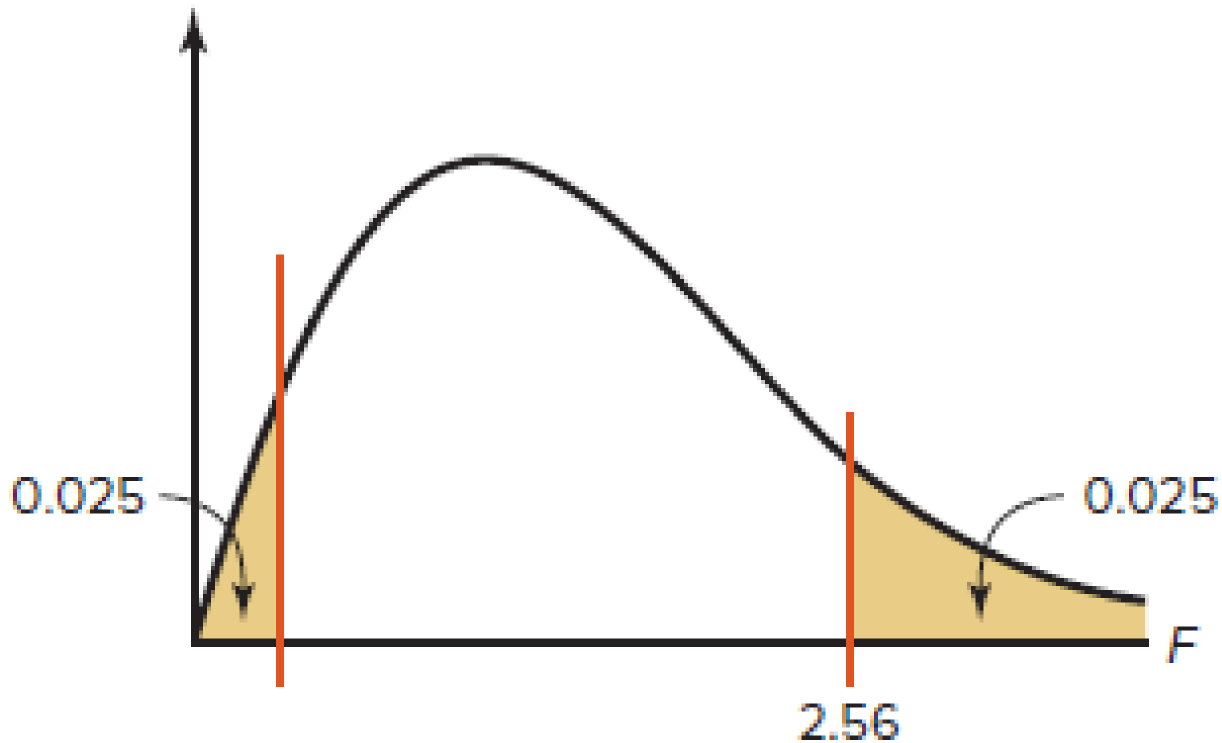
Step 3 Compute the test value.

$$F = \frac{s_1^2}{s_2^2} = \frac{36}{10} = 3.6$$

Step 4 Make the decision. Reject the null hypothesis, since 3.6 > 2.56.

Step 5 Summarize the results. There is enough evidence to support the claim that the variance of the heart rates of smokers and nonsmokers is different.

Ex 9.14 Critical Value & P-Value




```
> pf(3.6, 25, 17, lower.tail = FALSE)
[1] 0.004219627
> pf(2.548, 25, 17, lower.tail = FALSE)
[1] 0.02501926
```

Lecture Part II
R Practices

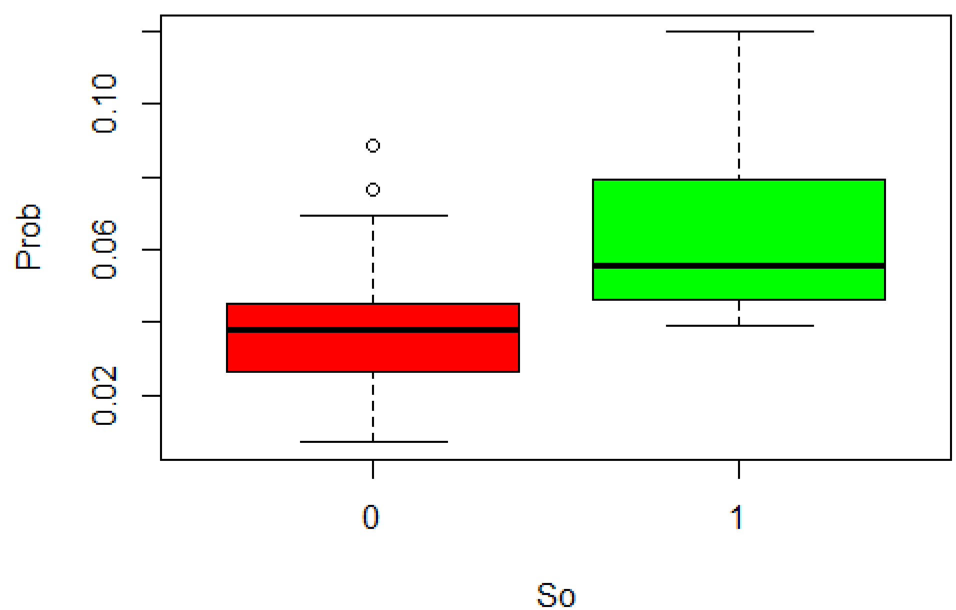
US Crime Data

- UScrime {MASS}

- The effect of punishment regimes on crime rates
  - So: binary indicator for a Southern state
  - Prob: probability of imprisonment
  - Po1: police expenditure in 1960
  - Po2: police expenditure in 1959

US Crime Data

```
> library(MASS)
> head(UScrime)
    M So  Ed Po1 Po2  LF  M.F Pop  NW  U1 U2 GDP Ineq     Prob    Time    y
1 151  1  91  58  56 510  950  33 301 108 41 394  261 0.084602 26.2011  791
2 143  0 113 103  95 583 1012  13 102  96 36 557  194 0.029599 25.2999 1635
3 142  1  89  45  44 533  969  18 219  94 33 318  250 0.083401 24.3006  578
4 136  0 121 149 141 577  994 157  80 102 39 673  167 0.015801 29.9012 1969
5 141  0 121 109 101 591  985  18  30  91 20 578  174 0.041399 21.2998 1234
6 121  0 110 118 115 547  964  25  44  84 29 689  126 0.034201 20.9995  682
```



Two Sample T-Test(Independent)

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

```
> aggregate(Prob ~ So, data = UScrime, FUN = mean)
  So       Prob
1  0 0.03851265
2  1 0.06371269
```

```
> t.test(Prob ~ So, data = UScrime)

	Welch Two Sample t-test

data:  Prob by So
t = -3.8954, df = 24.925, p-value = 0.0006506
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.03852569 -0.01187439
sample estimates:
mean in group 0 mean in group 1 
     0.03851265      0.06371269 
```

Two Sample T-Test(Dependent)

$$H_0\colon \mu_D = 0$$

$$H_1\colon \mu_D \neq 0$$

```
> sapply(UScrime[c("Po2", "Po1")], function(x)(mean=mean(x)))
     Po2      Po1
80.23404 85.00000
```

```
> with(UScrime, t.test(Po2, Po1, paried=TRUE))

        Welch Two Sample t-test

data:  Po2 and Po1
t = -0.80073, df = 91.66, p-value = 0.4254
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -16.587786   7.055871
sample estimates:
mean of x mean of y
 80.23404  85.00000
```

F-Test for Two Variances

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

```
> aggregate(Prob ~ So, data = UScrime, FUN = var)
  So         Prob
1  0 0.0003160508
2  1 0.0005064913

> 0.0003160508/0.0005064913
[1] 0.6240005
```

```
> var.test(Prob ~ So, data = UScrime)

        F test to compare two variances

data:  Prob by So
F = 0.624, num df = 30, denom df = 15, p-value = 0.2646
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.2360299 1.4396653
sample estimates:
ratio of variances
         0.6240006
```

```
> qf(0.025, df1=30, df2=15)
[1] 0.4334345
> qf(0.025, df1=30, df2=15, lower.tail = FALSE)
[1] 2.643735
```

Food for Thought

Will Sun Rise Tomorrow?

Image Credit: USA Today

Solution

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

θ: the probability of sun will rise

n: we have observed n times

k: and found k times of sun rise

$$f(\theta|nk) = \frac{f(nk|\theta)f(\theta)}{\int_0^1 f(nk|\theta)f(\theta)d\theta}$$

$f(\theta) = 1$: uniform distribution

$$f(\theta|nk) = \frac{f(nk|\theta)}{\int_0^1 f(nk|\theta)d\theta}$$

$$f(\theta|nk) = \frac{f(nk|\theta)}{\int_0^1 f(nk|\theta)d\theta}$$

Solution

$$f(nk|\theta) = \theta^k(1-\theta)^{n-k}$$

$$\int_0^1 \theta^k(1-\theta)^{n-k}d\theta = B(k+1, n-k+1)$$

$$f(\theta|nk) = \frac{(n+1)!}{k!\,(n-k)!}\theta^k(1-\theta)^{n-k}$$

$$E(\theta) = \int_0^1 \theta f(\theta|nk)d\theta$$

$$= \frac{(n+1)!}{k!\,(n-k)!}\int_0^1 \theta^{k+1}(1-\theta)^{n-k}d\theta$$

$$= \frac{(n+1)!}{k!\,(n-k)!}\frac{(k+1)!\,(n-k)!}{(n+2)!} = \boldsymbol{\frac{k+1}{n+2}}$$

99.989%

Homework

Task List

- View lessons in Canvas
- Read *Elementary Statistics*, Chapter 9
- Read *R in Action*, Chapter 7, 11.1
- Complete primary Discussion post by Thursday
- Complete Practice Problem Set (not submitted)
- Complete R Practice assignment
- Take Quiz
- Complete and submit Final Project - Milestone 2

What's Next
Correlation & Regression I

Questions?